Building an Intelligent Supply Chain Disruption Management Platform

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Overview

Your team at **GlobalLogic** has been hired by **NovaRetail Group**, a multinational retail and distribution organization, to build an AI-powered supply chain operations assistant.

NovaRetail manages suppliers, warehouses, shipments, inventory, and customer orders through multiple systems. When a shipment is delayed, inventory becomes unavailable, or a supplier fails to deliver, operations teams must manually check several portals and coordinate with different departments.

NovaRetail wants a single intelligent platform capable of understanding supply chain incidents, identifying the affected business area, invoking the appropriate operational tools, and coordinating recovery workflows.

The client expects a production-ready MVP within **two days**.

You will work as an AI consulting team, where each member is responsible for a different part of the solution.

Client Profile

Organization

NovaRetail Group

- 150 Retail Stores
- 8 Regional Warehouses
- 300+ Suppliers
- Nationwide Delivery Network
- Central Supply Chain Team
- 5,000 Daily Shipments

Existing Technology

- Supplier Management System
- Warehouse Management System
- Inventory Management System
- Shipment Tracking Platform
- Purchase Order System
- Customer Order Portal
- Notification System
- Email and Teams

All systems expose mock REST APIs or JSON data for this capstone. It will be provided with the project.

Current Business Challenges

Supply chain employees currently use multiple systems.

Examples:

- Track Shipments
- Check Inventory
- Find Supplier Details
- Identify Delayed Orders
- Report Damaged Goods
- Find Alternative Suppliers
- Create Supply Chain Incidents

Operations teams want one AI assistant instead of navigating several separate systems.

Executive Goals

The COO expects:

- Unified Operations Experience
- Intelligent Incident Routing
- Faster Disruption Response
- Reduced Manual Coordination
- Better Supply Chain Visibility

- AI-Assisted Recovery Planning

Project Objective

Build an AI-powered Supply Chain Assistant capable of understanding operational requests, deciding which supply chain function should handle them, executing the required workflow, and responding naturally.

Team Structure and Role

Students should work in teams of **4 members**.

Team Member 1

AI Conversation Engineer

Responsible for:

- Chat UI
- Prompt Engineering
- Memory
- Conversation History
- System Prompt
- User Experience

Team Member 2

Tool & Integration Engineer

Responsible for building tools such as:

- Shipment Lookup
- Inventory Checker
- Supplier Lookup

- Alternative Supplier Search
- Order Impact Checker
- Incident Creator
- Route Status Checker

Team Member 3

Agent Engineer

Responsible for:

- LangGraph Workflow
- Agent Logic
- Tool Selection
- Human-in-the-loop
- Error Handling

Team Member 4

Multi-Agent Engineer

Responsible for:

- Supervisor Agent
- Shipment Agent
- Inventory Agent
- Supplier Agent
- Recovery Agent
- Reporting Agent

All Members

Evaluation & Deployment Engineer

Responsible for:

- LangSmith Tracing
- Prompt Evaluation
- Testing
- Deployment
- Documentation

Functional Requirements

The assistant should support requests like:

Shipment

- Track Shipment
- Check Shipment Delay
- Identify Affected Orders

Inventory

- Check Product Stock
- Identify Inventory Shortage
- Check Warehouse Availability

Supplier

- Find Supplier Details
- Check Supplier Availability
- Find Alternative Suppliers

Incident Management

- Create Supply Chain Incident
- Check Incident Status
- Escalate Critical Incident

Recovery Planning

- Recommend Recovery Action

- Compare Supplier Alternatives
- Generate Incident Summary

Level 1

AI Assistant

Build a conversational supply chain operations assistant.

Features:

- Chat UI
- Memory
- Prompt Engineering
- Guardrails

Level 2

AI Agent

Convert the chatbot into an AI Agent capable of invoking tools.

Example tools:

```
track_shipment()

check_inventory()

search_supplier()

find_alternative_supplier()

check_affected_orders()

create_incident()
```

The agent should automatically determine which tool or tools to invoke.

Level 3

Multi-Agent Architecture

Build a LangGraph-based workflow.

Recommended Agents

Supervisor Agent

Receives every request.

Determines:

- Shipment
- Inventory
- Supplier
- Incident
- Recovery

Routes the request accordingly.

Shipment Agent

Handles:

- Shipment Tracking
- Delivery Delays
- Route Status
- Affected Orders

Inventory Agent

Handles:

- Product Availability
- Warehouse Stock
- Inventory Shortages

Supplier Agent

Handles:

- Supplier Information
- Supplier Availability
- Alternative Suppliers

Recovery Agent

Handles:

- Recovery Planning
- Alternative Recommendations
- Cost and Delay Comparison

Reporting Agent

Handles:

- Incident Summaries
- Operational Reports
- Stakeholder Updates

Human-in-the-Loop

The system must request confirmation before:

- Creating an Incident

- Selecting an Alternative Supplier
- Rerouting a Shipment
- Approving a Recovery Plan
- Sending a Stakeholder Notification

LangSmith Requirements

Every team must:

- Enable tracing
- Record at least 10 conversations
- Analyze latency
- Review failed runs
- Improve one prompt based on trace insights

Deployment

Deploy the application using:

- Streamlit
- Environment Variables
- Production-ready README

Deliverables

Each team must submit:

1. Working Application

Fully functional Streamlit application.

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2. Source Code

Well-structured repository with proper folder organization.

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3. Architecture Diagram

Illustrate:

- LLM
- Tools
- LangGraph Workflow
- Multi-Agent Design

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4. LangSmith Report

Include:

- Traces
- Prompt improvements
- Evaluation observations

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5. Deployment

Live URL using a free cloud service such as Streamlit Community Cloud, or a local deployment demonstration if internet access is unavailable.

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6. Technical Presentation

Include:

- Project Overview
- Setup Instructions
- Folder Structure
- Team Responsibilities
- Future Enhancements

PHASE 2 CLIENT ENHANCEMENT
CrisisOps AI
Version 2 – AI Supply Chain Digital Twin

Background

Following the success of CrisisOps AI, NovaRetail Group significantly improved its ability to respond to disruptions. However, executives identified another challenge: planners often need to choose among several recovery options without understanding their long-term operational or financial impact. Decisions such as changing suppliers, rerouting shipments or redistributing inventory are currently based on experience rather than simulation. The COO has initiated Phase 2 to introduce an AI-powered Supply Chain Digital Twin capable of evaluating alternative recovery strategies before execution.

Business Context

Global supply chains are highly interconnected. A single disruption can affect procurement, inventory, logistics, warehouse operations and customer commitments. Organizations increasingly use Digital Twins to simulate operational scenarios before making high-impact decisions. NovaRetail wants CrisisOps AI to support planners by comparing multiple recovery options and forecasting their consequences.

Phase 2 Client Requirement

Enhance CrisisOps AI with an AI-powered Digital Twin that models the current supply chain, simulates alternative disruption recovery strategies and recommends the most suitable approach based on operational objectives. The assistant should explain expected trade-offs, business impact and confidence before planners approve execution.

Illustrative Business Scenarios

- Compare rerouting shipments through two alternative distribution hubs.
- Evaluate replacing an unavailable supplier with two regional suppliers.
- Estimate inventory impact if warehouse capacity is temporarily reduced.

- Simulate demand spikes during a seasonal promotion.
- Compare transportation cost versus delivery time for multiple logistics partners.
- Assess the effect of delayed customs clearance on customer commitments.
- Estimate business impact of prioritizing premium customers during shortages.

Business Objectives

- Support simulation-based operational decision making.
- Compare multiple recovery strategies before execution.
- Reduce operational risk and unnecessary recovery costs.
- Improve planning transparency through explainable AI recommendations.
- Enable business users to evaluate trade-offs with confidence.

Architecture Exploration

Teams should design a Digital Twin architecture capable of representing suppliers, warehouses, transportation networks, inventory and customer demand. Students should explore how simulation, AI reasoning and operational constraints can be combined to compare alternative recovery plans. The implementation is intentionally open, allowing teams to justify different architectural approaches.

Design Challenges

- How should operational trade-offs be represented?
- Which business metrics should influence AI recommendations?
- How should simulation confidence be communicated?
- How can multiple recovery plans be compared fairly?
- What enterprise data should be synchronized with the Digital Twin?

Expected Deliverables

1. Extended architecture illustrating the Digital Twin layer.
2. Demonstration of five disruption simulation scenarios.
3. Design justification explaining simulation and recommendation strategy.
4. Discussion of scalability, assumptions and future enhancements.

Evaluation Expectations

Assessment will emphasize architectural reasoning, enterprise realism, quality of simulation thinking, explainability and the ability to justify business trade-offs. Different implementation strategies are acceptable provided they align with the business objectives.

Evaluation Criteria

Criteria	Weight(100)
Prompt Engineering & Conversation Design	15
LangChain & Tool Integration	20
Agent & LangGraph Workflow	20
Multi-Agent Collaboration	15
LangSmith Tracing & Evaluation	10
Deployment & Documentation	10
Team Collaboration & Final Presentation	10